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A deep learning–driven cyber attack avoidance framework for secure IoT-enabled smart city transportation systems

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Abstract

The widespread adoption of Internet of Things (IoT) technologies has transformed intelligent transportation systems (ITS) into a fundamental pillar of smart city infrastructures, enabling real-time traffic monitoring, connected vehicle communication, and adaptive mobility management over large-scale distributed networks. Despite these advancements, IoT-enabled transportation systems remain highly vulnerable to sophisticated cyber-attacks due to their decentralized architecture, heterogeneous devices, continuous data exchange, and stringent latency constraints. Existing cybersecurity solutions predominantly focus on reactive intrusion detection and post-incident mitigation, which are insufficient for mission-critical transportation services where even short disruptions can lead to severe safety and operational consequences. To address this challenge, this paper proposes a proactive deep learning–based cyber-attack avoidance framework tailored for IoT-enabled smart city transportation networks. The proposed approach models the transportation communication infrastructure as a dynamic graph and employs a hybrid spatio-temporal Graph Transformer architecture to jointly capture network topology–aware attack propagation patterns and long-range temporal dependencies in evolving cyber threats. A self-supervised pretraining strategy based on masked traffic modeling is incorporated to enhance robustness against zero-day and previously unseen attacks. Furthermore, a risk-aware decision layer is introduced to translate predicted attack trajectories into adaptive avoidance actions, including dynamic access control, intelligent routing adjustments, and selective node isolation, thereby preventing attack escalation before service disruption occurs. Experimental evaluation across UNSW-NB15, CICIDS2018, Bot-IoT, and Smart-ITS datasets demonstrate that the proposed framework consistently outperforms state-of-the-art models, achieving up to 98.64% accuracy, 98.04% F1-score, macro-F1 of 0.964, false alarm rates below 1%, and real-time response latency as low as 34.9 ms, while maintaining strong zero-day detection capability (AUROC = 0.963). The proposed framework delivers accurate, low-latency, and robust intrusion detection across diverse and large-scale

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cyber-physical environments, demonstrating its suitability for real-time deployment in next-generation intelligent networks.

Keywords Internet of things (IoT) · Security intelligent transportation systems · Cyber attack avoidance · Spatio-temporal graph transformer · Zero-day attack detection · Smart city cybersecurity

1 Introduction

The rapid evolution of smart cities has been fundamentally driven by the widespread deployment of Internet of Things (IoT) technologies across critical urban infrastructures, with intelligent transportation systems (ITS) occupying a central and safety–critical role within this ecosystem [1]. The IoT-based transportation systems combine heterogeneous sensors, roadside units, connected vehicles, traffic controllers, and cloud-edge computing platforms to allow real-time traffic monitoring and adaptive signal control, cooperative vehicle-infrastructure communication, and data-driven mobility optimization. The capabilities play a critical role in improving the efficiency of traffic, decreasing congestion, minimizing emissions, and enhancing road safety [2]. The same features that enable intelligent transportation—such as large-scale connectivity, continuous data exchange, decentralized control, and stringent low-latency requirements—also expand the cyberattack surface of smart city transportation networks, thereby making cybersecurity a critical challenge for next-generation urban mobility systems.

The IoT-based transportation systems communication structure is complex and dynamic in nature [3]. Traffic lights, vehicle onboard units, cameras, and control centres communicate via wireless and wired networks with varied communication protocols and data formats. This heterogeneity, together with the limited resources of the devices and the constant changes of topology because of the mobility of the vehicles, introduces vulnerabilities at the systemic level that adversaries can exploit [4]. The types of cyber-attacks against transportation infrastructure can be denial of service attacks on traffic controllers, spoofing vehicle messages, manipulation of data in traffic sensors, malware used to spread across roadside units, and coordinated attacks based on the interdependence of network components [5]. In contrast to traditional information systems, cyber-attacks in transportation networks can instantly be converted into physical impacts, including gridlocks, crashes, delays in emergency response, and cascading failures of interconnected urban services.

The conventional cybersecurity systems used in transportation systems have mostly been borrowed from enterprises or general IoT security paradigms [6]. The approaches are usually based on perimeter-based defenses, signature-based intrusion detection systems, anomaly detection by rules, and centralized monitoring architectures. Although these methods have been effective in dealing with known threats, they are basically reactive in nature [7]. They do not identify attacks until the malicious activity has occurred, and in many cases, they cause mitigation measures when the system has already been partially compromised. Reactive defense strategies are

inadequate in transport systems where the delay of a millisecond or continuous availability is critical to the mission. Any temporary inconveniences or slow reactions can undermine the safety, destroy the confidence of the population, and lead to serious financial losses.

The recent developments in machine learning and deep learning have sparked the desire to move towards intelligent cybersecurity, which can teach intricate attack patterns based on data. The use of supervised and unsupervised learning models has been implemented in intrusion detection in vehicular networks, traffic management systems, and smart city IoT infrastructures. Deep neural networks, recurrence-based, and autoencoder-based models are better detectors than traditional approaches because they can capture nonlinear relationships in network traffic and system behavior [8]. The majority of learning-based solutions available are still concentrated on attack detection as opposed to attack avoidance. They strive to categorize traffic as either benign or malicious, without necessarily modeling the way attacks change with time or how defensive measures are to be taken in advance to avoid escalation. Most of the models assume network traffic to be independent samples, ignoring the graph nature of transportation communication networks and the spatio-temporal dependencies that regulate attack propagation [9].

This study is motivated by the urgency to go beyond reactive intrusion detection to proactive cyber-attack prevention in the IoT-based smart city transportation networks. Attack avoidance focuses on predicting malicious behavior prior to its impact on the availability and safety of a service and using this prediction to adaptively reconfigure the system. This goal needs a cohesive framework that is capable of (i) modeling the transportation communication infrastructure as a dynamic, interconnected system, (ii) learning new patterns of attacks as they evolve under highly imbalanced and non-stationary traffic conditions, (iii) generalizing to new and novel attacks, and (iv) converting predictive information into timely, risk-sensitive defensive measures. The current methods are not fully satisfying these needs, and thus, they end up providing solutions that are not complete, and thus, they are hard to implement in the real-world smart city settings.

One of the research gaps that requires critical research is the scarcity of exploiting the use of graph-based deep learning to provide proactive cybersecurity in transportation networks. Although graph neural networks have been studied in network security, their use in smart transportation is sparse and usually static, and cannot be used to view long-range temporal interactions and dynamical topological interactions. In the same way, transformer-based architecture, which has shown great performance in the modeling of sequential and relational information, has not been adequately customized to the combined spatio-temporal properties of the cyber threats in IoT transportation systems. The other notable gap is that there is no self-supervised learning strategy that is specific to transportation traffic, which can decrease the reliance on labeled attack data and enhance resistance to new threats. Moreover, the majority of the existing literature ends up with the metrics of detection performance, and there is little discussion about how predictive models can be used to inform the operational decisions actively preventing the escalation of attacks.

In this regard, the Research Questions (RQs) that the paper dealing with can be formulated as follows: How can a smart city transportation system proactively

anticipate and prevent cyber-attacks in a highly dynamic, heterogeneous IoT environment, without compromising on low latency, high availability, and operational safety? This general issue includes a number of interconnected challenges, such as proper modeling of spatio-temporal attack propagation, early detection of malicious paths, managing data scarcity and concept drift, and formulating decision processes that are sensitive to security and transportation performance issues.

The primary objective of this study is to design and evaluate a proactive deep learning–driven cyber-attack avoidance framework that enhances the security, resilience, and reliability of IoT-enabled smart city transportation systems by predicting and mitigating cyber threats before service disruption occurs. The major contributions of this study are as follows:

- A proactive cyber-attack avoidance framework is proposed for IoT-enabled smart city transportation systems, shifting the focus from reactive intrusion detection to predictive and preventive cybersecurity.
- A hybrid spatio-temporal Graph Transformer architecture is designed to jointly model network topology–aware attack propagation and long-range temporal dependencies in evolving transportation traffic.
- It presents a self-supervised pretraining approach, which is called masked traffic modeling and is used to enhance resistance to zero-day attacks and novel cyber-attacks.
- A risk-aware decision layer is created to transform the forecasted attack paths into dynamic avoidance responses, such as dynamic access control, smart routing modifications, and node isolation.
- Comprehensive experimental testing on benchmark four intrusion datasets and simulated smart transportation traffic conditions shows better performance in accuracy of prediction, false alarm reduction, response latency, and system availability than current machine learning and deep learning techniques.

The rest of the paper is structured in the following way. Section 2 conducts a review of related literature in the field of cybersecurity in IoT-enabled transportation systems, deep learning-based intrusion detection, and graph-based security modeling, all of which are limited in their ability to prevent attacks proactively. Section 3 gives the system model and problem formulation, and then gives a detailed description of the proposed deep learning-based attack avoidance framework. Section 4 then discusses the experimental setup, datasets, evaluation metrics, and comparative results with a view to confirming effectiveness of the proposed approach. The discussion is given in Sect. 5. Section 6 is a conclusion of the paper with a certain scope of future research.

2 Literature review

Recent advances in smart cities and intelligent transportation systems have significantly increased reliance on IoT-enabled, data-driven infrastructures, thereby expanding the cyber-attack surface and exposing transportation services to

sophisticated and evolving threats. Consequently, a growing body of literature has explored machine learning and deep learning–based security mechanisms for intrusion detection, anomaly analysis, and cyber-defense in transportation and urban IoT environments. This section reviews and critically analyzes existing state-of-the-art approaches.

Maranco et al. [10] proposed an all-encompassing conceptual framework of cybersecurity of Intelligent Transportation Systems (ITS) that incorporates the assessment of risks, real-time threat detection, incident response, and continuous monitoring based on AI/ML. Their work identified important vulnerabilities that occur due to the integration of IoT and real-time data dependency in ITS architecture. The research is mostly theoretical, without empirical confirmation, quantitative performance analysis, and practical implementation examination.

Kohli et al. [11] suggested an AI-based Detection System (IDS) called Multi-Branch Reconstruction Error (MbRE), which is trained on the normal behavior of vehicles and identifies abnormalities at edge nodes. The method showed a very high recall and accuracy of between 97.5 and 100 percent of known and unknown attacks. Although it performed well, it was only tested in controlled experimental conditions, without considering large-scale implementation, computational complexity, or scalability in the long term.

Maqousi et al. [12] introduced an independent AI-based cyber-defense system that focuses on learning and real-time analytics of smart city systems. The framework was effective in decreasing the detection to response time and was illustrated with representative case studies. Lack of standardized benchmarks and quantitative comparative analysis does not allow objectively evaluation of its performance.

The model of multi-layer cybersecurity that was created by Mustapha and Alabi et al. [13] includes machine-learning-based anomaly detection, blockchain-based threat intelligence sharing, and a zero-trust architecture. Their strategy enhanced response time and accuracy of detection against ransomware, DDoS, and advanced persistent threats. The research fails to critically examine the complexity of implementation, large ITS deployment scalability, and runtime overhead.

Singh et al. [14] introduced a Multi-Classes Intrusion Detection System (MCIDS) that is built on deep-learning and trained on the UNSW-NB15 dataset to detect various types of IoT attacks. The system was very accurate and had low false positive rates. They were only validated using one benchmark dataset and had not investigated generalization to heterogeneous IoT traffic and real-time operational settings.

Alrashdi et al. [15] proposed an anomaly detection framework (AD-IoT) based on the framework of the Random Forest and implemented it on the fog nodes in smart cities. The system had a high accuracy of classification of 99.34 per cent and a low false positive rate. Although these are encouraging outcomes, flexibility to various smart-city systems and changing cyberattack trends is yet to be dealt with.

Alkhudhayr and Ardah et al. [16] suggested an IDS using machine-learning to identify intrusion in self-driving car cyber-physical systems (CPS) using krill-herd-optimized decision trees. The method was able to detect cyber-physical attacks with 98.3 percent accuracy, only in simulated settings, without taking into account the problem of deploying the sensor in the real world or sensor-level uncertainties.

Zhu et al. [17] paid attention to the development of online gray-box data-poisoning attacks on edge AI systems by gradient-based and loss-based approaches. Their findings showed that they were able to poison online with minimal computational cost. The paper focuses on the effectiveness of attacks and never explores defensive mechanisms and resilience strategies of edge AI systems.

Aleisa et al. [18] designed an ensemble LSTM-based IDS to Intelligent Cyber-Physical Transportation Systems (ICTS), which tracks in-vehicle networks, V2V, and V2I communications. The suggested system was 100 percent accurate on the car-hacking dataset and 99 percent accurate on UNSW-NB15. Although it had high detection performance, it was not discussed in terms of computational complexity, latency constraints, and feasibility on resource-constrained vehicular platforms.

Qu et al. [19], examined the adversarial collusion attack on deep reinforcement learning-based traffic signal control systems by the CollusionVeh framework. Their results indicated that colluding vehicles could save a lot of time by falsifying traffic data and reducing travel time. The study mainly focuses on attack modeling and does not suggest mitigation strategies or secure reinforcement learning defenses.

Dayoub et al. [20] proposed a conceptualization of machine learning, blockchain, secure routing, and AI to improve cybersecurity and privacy of urban IoT systems. The article highlights the significance of the use of AI in security by smart cities. The lack of experimental validation limits the evaluation of practical effectiveness and scalability.

Duclos et al. [21] proposed a reinforcement-learning-based cybersecurity framework for Autonomous Intersection Management (AIM) systems using PPO and A2C algorithms. The framework demonstrated adaptive threat detection and mitigation in simulation-based digital twin environments. Real-world intersection data, sensor noise, and operational constraints were not evaluated.

Hatzivasilis et al. [22] introduced SPD-Safe, an AI-enabled security framework for railway cyber-physical systems, integrating secure routing and authentication mechanisms. The framework improved railway CPS security with low operational latency. Interoperability with external emergency-response systems remains an unresolved challenge.

Almasri and Alajlan et al. [23] proposed a cascaded ANFIS combined with modified deep reinforcement learning for cyberattack detection and isolation in cloud IoT environments. The system achieved approximately 98.7% detection accuracy and effectively isolated compromised devices. Computational overhead, convergence stability, and scalability in dense IoT networks were not analyzed.

Reis et al. [24] developed a hybrid deep learning IDS combining autoencoders, CNNs, and LSTM networks with federated learning and edge AI. The framework had high precision, recall, and F1 scores, and also facilitated privacy-preserving distributed learning, communication overhead and synchronization issues related to federated learning had not been quantitatively evaluated.

The TSA-optimized stacked sparse autoencoder (TSA-OSSAE) proposed by Das et al. [25] is used to detect cyber threats in IoT-enabled smart cities. The approach performed better than the baseline IDS approaches on the TON IoT dataset. Assessment was not done on multiple datasets, and real-time integration of the system was not examined.

Nair et al. [26] suggested an interpretable deep-learning IDS that can be used to understand transportation IoT security through the incorporation of SHAP explanations. The model was very accurate (99.15) and enhances transparency in intrusion decisions. There were no trade-offs studied between interpretability, latency in detection, and scalability.

Saba et al. [27] utilized a deep learning IDS that had a voting classifier with several TON_IoT datasets with an accuracy of 99.7%. Although performance was good, resistance to concept drift and changing strategies to counter cyberattacks over time was not studied.

Bergies et al. [28] suggested a model predictive control, deep neural networks, dynamic programming, and homomorphic encryption-based IoT-based architecture to secure automated electric vehicles. The framework increased security and maintained the privacy of data. The computational cost of homomorphic encryption in real-time vehicular systems was not measured.

The article by Nagarajan et al. [29] introduced a deep learning IDS with feature selection and an enhanced RSA-based secure transmission scheme. The solution enhanced the precision of intrusion detection and safety in the delivery of data. Scalability to heterogeneous IoT systems and scalability to large-scale implementations are open research problems. Table 1 provides a summary of the reviewed literature.

The literature review above indicates that there are a number of systematic research gaps in the current body of knowledge on cybersecurity of IoT-enabled intelligent transportation systems and smart city systems [30]. Most of the previous research is on reactive intrusion detection as opposed to proactive attack avoidance, where the threats are detected after the malicious activity has already affected the system. Conceptual frameworks and AI-based defense architectures focus on detection and response but do not have mechanisms to predict the development of attacks and avoid escalation before the service is disrupted [30]. Most solutions are very accurate in detection in controlled or simulated conditions, but fail to deal with the real-world implementation issues, such as large-scale scalability, mixed traffic patterns, dynamic network topologies, and strict latency requirements of the transportation systems. Many of the works are based on single datasets or isolated benchmarks, which restrict cross-domain generalization and resilience to a wide range of attack surfaces. Computational overhead, response time, and practicability on resource-limited vehicle and edge platforms are not given much consideration, yet they are essential to safety–critical ITS functionality. Defensive resilience in the case of zero-day attacks and concept drift is not well studied, whereas some works investigate more sophisticated attack models, like data poisoning or adversarial collusion. The currently available solutions seldom incorporate risk-aware decision-making, and the results of detection are directly converted into adaptive control and mitigation measures that maintain system availability. The proposed deep learning-based cyber-attack avoidance system directly fills these gaps with the help of a single, proactive, and deployment-focused architecture. In contrast to the classical IDS-based solutions, the suggested approach explicitly represents attack paths on dynamic spatio-temporal graphs, which allows predicting the propagation of attacks early and preventing attacks, instead of applying mitigation measures after the incident. The combination of a spatio-temporal Graph Transformer captures the topology-based

Table 1 Summary of existing cybersecurity methods for intelligent transportation and smart city IoT systems

References	Methods	Key findings	Limitations
Maranco et al. [10]	Conceptual cybersecurity framework for ITS integrating risk assessment, threat detection, incident response, and AI/ML-based continuous monitoring	Identified critical ITS vulnerabilities due to IoT integration and real-time data reliance; proposed a holistic protection architecture	The study primarily presents a conceptual framework; empirical validation, quantitative performance evaluation, and real-world deployment analysis are not provided
Kohli et al. [11]	AI-based MbRE IDS using normal-behavior learning and deviation analysis at edge nodes	Achieved 97.5–100% recall and accuracy; effective for known and unknown attacks	Evaluation is conducted under controlled experimental settings; large-scale real-world deployment, computational overhead at edge nodes, and long-term adaptability are not examined
Maqousi et al. [12]	Autonomous AI-driven cyber-defense framework with continuous learning and real-time analytics	Reduced detection-to-response time; demonstrated effectiveness via case studies	The framework is validated through illustrative case studies; the absence of standardized benchmarks and quantitative comparative analysis limits objective performance assessment
Mustapha & Alabi, et al. [13]	Multi-layer security model with ML-based anomaly detection, blockchain-based threat intelligence, and zero-trust architecture	Improved detection accuracy and faster response to ransomware, DDoS, and APTs	While the model integrates multiple security layers, implementation complexity, scalability in large ITS environments, and runtime overhead are not systematically analyzed
Singh et al. [14]	Deep-learning-based Multi-Classifier IDS (MCIDS) using UNSW-NB15 dataset	High accuracy and low false positives across multiple IoT attack categories	The approach is validated on a single benchmark dataset; generalization to heterogeneous IoT traffic patterns and real-time deployment constraints is not explored
Alrashdi et al. [15]	Random Forest–based anomaly detection deployed at fog nodes (AD-IoT)	Achieved 99.34% accuracy with low false positives	The system is evaluated using a specific dataset and fog-based setup; adaptability to diverse smart-city infrastructures and evolving attack patterns remain unaddressed

Table 1 (continued)

References	Methods	Key findings	Limitations
Alkhudhayr & Ardah, et al. [16]	ML-based IDS for self-driving car CPS with krill-herd-optimized decision trees	Achieved 98.3% detection accuracy for cyber-physical attacks	The proposed IDS is tested in simulation environments; real-world vehicular CPS deployment challenges and sensor-level uncertainties are not considered
Zhu et al. [17]	Online gray-box data-poisoning attacks for edge AI using gradient- and loss-based strategies	Demonstrated effective online poisoning with low overhead	The study focuses on attack formulation and effectiveness; defensive strategies, detection countermeasures, and resilience of edge AI systems are outside the scope
Aleisa et al. [18]	Ensemble LSTM-based IDS for ICTS monitoring IVN, V2V, and V2I communications	Achieved 100% accuracy (car-hacking dataset) and 99% (UNSW-NB15)	High detection accuracy is reported; however, computational complexity, latency constraints, and feasibility for resource-constrained vehicular platforms are not discussed
Qu et al. [19]	Adversarial collusion framework (CollusionVeh) attacking DRL-based traffic signal control	It showed that colluding vehicles can reduce travel time by falsifying data	The work emphasizes adversarial attack modeling; mitigation mechanisms and secure reinforcement learning defenses are not investigated
Dayoub, et al. [20]	Conceptual integration of ML, blockchain, secure routing, and AI for urban IoT resilience	Highlighted the importance of AI-driven cybersecurity for smart cities	The paper adopts a conceptual and integrative perspective; the lack of experimental validation limits assessment of practical effectiveness and scalability
Duclos et al. [21]	Reinforcement Learning–based cybersecurity framework using PPO and A2C in AIM systems	Demonstrated adaptive threat detection and mitigation in simulations	The framework relies on simulation-based digital twins; real-world intersection data, sensor noise, and operational constraints are not evaluated
Hatzivasilis et al. [22]	SPD-Safe AI-enabled framework for railway CPS with secure routing and authentication	Improved railway CPS security with low operational latency	Although validated in controlled deployments, seamless interoperability with external emergency-response systems remains an open challenge

Table 1 (continued)

References	Methods	Key findings	Limitations
Almasri & Alajlan, et al. [23]	Cascaded ANFIS with modified deep reinforcement learning for attack detection and isolation	Achieved ~98.7% accuracy; effective isolation of compromised devices	The approach demonstrates high accuracy; however, computational overhead, convergence stability, and scalability in dense IoT networks are not analyzed
Reis, et al. [24]	Hybrid deep learning IDS (Autoencoder + CNN + LSTM) with federated learning and edge AI	Achieved high precision, recall, and F1; scalable and privacy-preserving	Federated learning introduces communication overhead and synchronization challenges, which are not quantitatively assessed in large-scale deployments
Das et al. [25]	TSA-optimized stacked sparse autoencoder (TSA-OSSAE) for cyber threat detection	Outperformed baseline IDS methods on the TON_IoT dataset	Evaluation is limited to a single telemetry dataset; cross-dataset generalization and real-time system integration are not explored
Nair et al. [26]	Interpretable deep learning IDS using SHAP for transportation IoT security	High accuracy (99.15%) with explainable intrusion decisions	While interpretability is addressed, trade-offs between explainability, detection latency, and system scalability are not investigated
Saba et al. [27]	Deep learning IDS with voting classifier using multiple TON_IoT datasets	Achieved 99.7% accuracy across several IoT device datasets	The IDS achieves high accuracy; however, robustness against concept drift and evolving attack strategies over time is not examined
Bergies et al. [28]	IoT-based architecture integrating MPC, DNN, DP, and homomorphic encryption	Enhanced AEV security while preserving data privacy	Privacy-preserving mechanisms are incorporated; nevertheless, the computational cost of homomorphic encryption in real-time vehicular systems is not quantified
Nagarajan et al. [29]	Deep learning IDS with feature selection and improved RSA-based secure transmission	Improved detection accuracy and secure data delivery	The study demonstrates improved detection accuracy; adaptability to heterogeneous IoT ecosystems and large-scale deployment feasibility remain unaddressed

dependencies, as well as long-range temporal patterns, going beyond the constraints of the non-dynamic graphs and strictly sequential models in a dynamic ITS context.

3 Materials and methods

Figure 1 provides an overview of the proposed approach. The suggested solution proposes a proactive, and predictive model of cybersecurity of IoT-enabled smart city transportation systems that fundamentally changes the defensive strategies by focusing on intrusion detection response to early attack prevention. The framework starts by simulating the heterogeneous transportation cyber infrastructure as a time-varying attributed graph, with vehicles, roadside units, IoT sensors, and control servers being the nodes, and time-varying communication interactions being directed edges. This formulation allows a clear representation of both the spatial connectivity and temporal evolution, which are crucial in comprehending the emergence, propagation, and escalation of cyber threats across the interconnected ITS components. Raw traffic, protocol, and security data are systematically preprocessed and encoded to build semantically meaningful node representations, and temporal aggregation reduces noise and adopts new attack signatures. The paradigm is centered around a Spatio-Temporal Graph Transformer, which concurrently learns the topology-aware spatial dependencies and the long-range temporal dependencies by using attention mechanisms. The architecture also allows the model to discover both important communication paths over which attacks propagate and also to detect early-stage, stealthy, or multi-step attack patterns that are frequently overlooked by other sequence models. To become more resilient to zero-day and never-before-seen attacks, a self-supervised masked traffic modeling approach is incorporated to enable the framework to acquire intrinsic traffic semantics and generalized behavioral

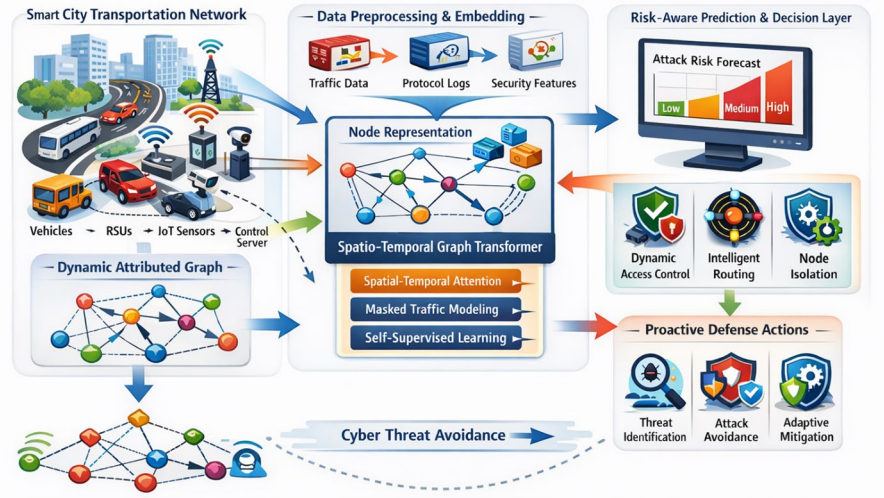


Fig. 1 Overview of the proposed approach

patterns without necessarily using labeled attack data. A risk-aware prediction and decision layer is used to convert probabilistic attack predictions to adaptive avoidance responses, including dynamic access control, intelligent routing, and selective node isolation, to avoid escalation of an attack before service disruption ensues.

3.1 Problem formulation and notation

A smart city transportation system that is based on an IoT is intrinsically defined by massive, heterogeneous, and dynamically changing communication interactions between sensing units, roadside infrastructure, vehicular nodes, and control servers. To model this complexity formally, the transportation cyber infrastructure is represented as a dynamic attributed graph (discrete-time). At each time step $t \in \{1,2,\ldots,T\}$, the system state is represented as

$$\mathcal{G}_t = \left(\mathcal{V}_t, \mathcal{E}_t, X_t\right), \tag{1}$$

where $\mathcal{V}_t = \{v_1, v_2, \ldots, v_{N_t}\}$ denotes the set of IoT entities,$\mathcal{E}_t \subseteq \mathcal{V}_t \times \mathcal{V}_t$ represents directed communication links, and $X_t \in \mathbb{R}^{N_t \times d}$ is the node attribute matrix. This representation represents a graph signal, which is time-indexed, and learning is conducted on it. It allows modeling the propagation of cyber-attacks explicitly, both spatially and overtime.

All node-level feature vectors $x_{i,t} \in \mathbb{R}^d$ are built by concatenating traffic-level statistics, protocol semantics, and security-related indicators:

$$x_{i,t} = \left[x_{i,t}^{traf}, x_{i,t}^{proto}, x_{i,t}^{sec}\right]. \tag{2}$$

This decomposition divides the feature space into semantically meaningful subspaces. It maintains the complex character of cyber threats to transportation networks, where the attacks are observed at the same time in traffic anomalies, protocol abuse, and security breaches.

The learning goal is to predict the threats of future attacks using the past graph sequences. This is formalized by learning a parametric function.

$$f_\Theta : \left\{\mathcal{G}_{t-T+1}, \ldots, \mathcal{G}_t\right\} \rightarrow \hat{y}_{t+1}, \tag{3}$$

where $\hat{y}_{t+1} \in [0,1]^{N_t}$ represents node-wise risk scores of attacks, and Θ the parameters to be trained. This mapping is a mathematical formalization of proactive attack avoidance; it conceptually transforms the concept of defense into the concept of prediction.

3.2 Dataset

3.2.1 UNSW-NB15 dataset

UNSW-NB15 Dataset [31] is a popular dataset used to test the workability of intrusion detection and cybersecurity models in general IoT and networked settings. It

includes 257,673 records of network traffic modeled by 49 features that are carefully engineered and nine attack types, such as Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms, and normal traffic. The feature set is a combination of flow-based features (e.g., dur, spkts, dpkts, sbytes, dbytes, rate, sload, dload) and content- and state-sensitive features (e.g., proto, service, state, sttl, dttl, tcprtt, synack, ackdat). Besides that, the dataset contains connection- and behavior-level statistics, including ct_srv_src, ct_state-ttl, ct_dst-ltm, ct-src-dport-ltm, and ct-dst-src-ltm, that reflect the temporal dependencies and session reuse patterns. Application-level indicators, such as is_ftp_login, ct_ftp_cmd, ct_flw_http_mthd, and is_sm_ips_ports, add to the semantic characterization of attacks even more.

Dataset Link: https://research.unsw.edu.au/projects/unsw-nb15-dataset.

3.2.2 CICIDS2018 dataset

The CICIDS2018 dataset [32] is one of the largest and most inclusive benchmarks of contemporary cyber-attack detection that includes a set of more than 16 million network flow records with 80 high-dimensional features generated via the CICFlowMeter tool. The data set comprises 14 types of attacks, such as brute-force attacks, DDoS attacks, botnet activity, web-based attacks, infiltration, and benign traffic, and thus covers modern threat environments. Its characteristics are mostly flow-statistical and temporal, including Flow Duration, Total Fwd Packets, Total Bwd Packets, Flow Bytes/s, and Flow Packets/s, which are used to measure the volume and intensity of traffic. Inter-packet timing behavior is captured by fine-grained temporal descriptors, such as Flow IAT Mean, Flow IAT Std, Fwd IAT Mean, and Bwd IAT Mean. Protocol misuse and attack signatures can be understood by the use of packet-level statistics like Fwd Packet Length Mean, Bwd Packet Length Mean, Packet Length Variance, Avg Packet Size, and TCP flag counts (SYN, ACK, PSH).

Dataset Access: https://www.unb.ca/cic/datasets/ids-2018.html.

3.2.3 Bot-IoT dataset

The Bot-IoT dataset [33] is tailored to the IoT-focused model of botnets and massive automated attacks, and it is of high interest to the study of the smart city and industrial IoT security. It has over 72 million traffic instances, 46 features used to describe it, and 10 attack types, such as DDoS, DoS, reconnaissance, data exfiltration, and normal IoT behavior. The data set focuses on lightweight traffic statistics and response, i.e., pkSeqID, proto, saddr, daddr, sport, and dport, as the IoT devices are limited. Packet sequencing and traffic burst patterns, such as those in botnet activity, are described by temporal and statistical features such as seq, stddev, min, max, mean, srate, and drate. Other attributes like N_IN_Conn_P_SrcIP, state_number, and flgs_number contain the density of connections and protocol state transitions, which is important in detecting coordinated attacks that may have started at compromised IoT nodes.

Dataset Access: https://www.unsw.adfa.edu.au/unsw-canberra-cyber/cybersecurity/ADFA-NB15-Datasets/bot_iot.php.

3.2.4 Smart-ITS simulated traffic dataset

The Smart-ITS simulated traffic data [34] is specific to the vehicular IoT setting, and it involves the communication patterns in the Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) contexts of an intelligent transportation system. It is made up of about 1.2 million samples, each of which is represented by 52 ITS-specific features, and seven types of attacks, as well as normal vehicular communication. This dataset is a combination of mobility-, topology-, and context-aware data, unlike traditional network data sets, including vehicle-id, rsu-id, link-density, avg-speed, and hop-count, which characterize dynamic road-network interactions. The network performance indicators, such as packet_delay, packet_loss, jitter, and flow_duration, represent latency-sensitive communication needs of safety–critical ITS applications. Spatial connectivity and the change of traffic over time are explicitly represented in the graph and topology-related features, including node_degree, betweenness, route_entropy, and temporal flow-rate. Wireless channel effects are also added to signal-level attributes such as signal strength.

Simulation Tools & Datasets: https://www.eclipse.org/sumo/ https://veins.car2x.org/.

Note: Smart-ITS datasets are typically generated using validated vehicular simulators (SUMO, Veins, OMNeT++) and are widely accepted in ITS security literature for V2V/V2I attack modeling.

3.3 Data preprocessing and feature engineering

Raw traffic characteristics are highly scaled with stochastic noise. Thus, all continuous variables are normalized with a z-score:

$$\widetilde{x}_{i,t} = \frac{x_{i,t} - \mu}{\sigma}, \tag{4}$$

where μ and σ denote empirical mean and standard deviation vectors. This transformation enforces zero-mean and unit variance. In Eq. (4), $x_{i,t}$ denotes the original value of the ith continuous traffic feature observed at time step t, while $\widetilde{x}_{i,t}$ represents its normalized counterpart obtained through z-score standardization. The parameters μ and σ correspond to the empirical mean and standard deviation vectors computed across the training dataset for each continuous feature dimension.

Categorical attributes on a protocol level are converted to dense embeddings through Eq. (5).

$$e_{i,t} = W_e \bullet onehot\left(x_{i,t}^{proto}\right), \tag{5}$$

In Eq. (5), $x_{i,t}^{\text{proto}}$ refers to the categorical protocol-level attribute (e.g., TCP, UDP, ICMP) associated with the i-th traffic instance at time t. The operator onehot$(\cdot)$ converts this categorical variable into a one-hot encoded vector, which is then projected into a dense, low-dimensional embedding space through multiplication with the learnable embedding matrix W_e. The resulting vector $e_{i,t}$ captures latent semantic

relationships among protocol types, enabling the model to learn protocol-specific behavioral patterns that are critical for identifying protocol-oriented and stealthy cyber exploits. This allows the incessant representation learning of discrete protocol semantics that are vital in detecting protocol-based exploits.

Temporal aggregation is performed using sliding windows to mitigate burst noise using Eq. (6):

$$x_t = \frac{1}{|\mathcal{W}_t|} \sum_{k \in \mathcal{W}_t} \widetilde{X}_k, \tag{6}$$

In Eq. (6), $\widetilde{X}_k$ denotes the normalized feature vector at time index k, and W_t represents the set of indices within a sliding temporal window centered at time t. The window size $| W_t |$ defines the number of consecutive samples used for aggregation. The aggregated feature representation x_t is obtained by computing the average of normalized features within this window, effectively smoothing burst noise and short-term fluctuations in traffic.

Network topology is encoded through an adjacency matrix as presented in Eq. (7):

$$A_j(i,j) = \begin{cases} 1, & (v_i, v_j) \in \mathcal{E}_t \\ 0, & otherwise. \end{cases} \tag{7}$$

This clearly stipulates the flow paths of information that are allowed in relation to the spread of attacks. In Eq. (7), the network topology at time t is represented using an adjacency matrix A_t, where each element $A_t(i,j)$ defines the existence of a communication or dependency relationship between nodes v_i and v_j. The $A_t(i,j) = 1$ indicates that a direct edge exists between node v_i and node v_j, meaning that traffic flows or interactions are permitted between these entities at time t. Conversely, $A_t(i,j) = 0$ signifies the absence of a direct connection, implying no immediate information exchange between the corresponding nodes. The set E_t denotes the collection of active edges in the network graph at time t, capturing dynamic connectivity patterns that may evolve due to mobility, routing changes, or attack-induced disruptions.

3.4 Spatio-temporal graph transformer architecture

The framework consists of a Spatio-Temporal Graph Transformer (ST-GT) [35] that combines graph attention with temporal self-attention to simultaneously represent spatial dependencies and long-term temporal correlations. In the spatial dimension, the multi-head graph attention layers are dynamically trained to understand the significance of adjacent nodes and communication links so that the model can prioritize the critical pathways in which the attacks can propagate. The attention mechanism is dynamic and can be tailored to the changing conditions of the network, unlike the static graph convolution, which is appropriate in highly dynamic ITS settings. Stacked Transformer blocks in the temporal dimension can learn dynamic attack patterns across long temporal scales, allowing the model to identify early-stage attacks, slow reconnaissance, and multi-step attacks, which are challenging to

detect with short-term temporal models like RNNs or LSTMs. Figure 2 shows the summary of the proposed ST-GT architecture.

The features of the initial nodes are mapped into a latent space with the help of Eq. (8).

$$H_t^{(0)} = X_t W_0, \tag{8}$$

where $W_0 \in \mathbb{R}^{d \times d_h}$. this linear transformation aligns heterogeneous inputs. It prepares features for unified spatio-temporal reasoning.

Graph transformer attention is used to model spatial dependencies and calculates attention coefficients in the form of Eq. (9).

$$\alpha_{ij}^{(l)} = \frac{\exp\left(\frac{\left(Q_i^{(l)} K_j^{(l)^T}\right)}{\sqrt{d_h}}\right)}{\sum_{k \in \mathcal{N}(i)} \exp\left(\frac{\left(Q_i^{(l)} K_j^{(l)^T}\right)}{\sqrt{d_h}}\right)}. \tag{9}$$

This equalizes the scores of neighbor influences. It gives precedence to critical links of communication that aid in the propagation of attacks. In Eq. (9), $\alpha_{ij}^{(l)}$ denotes the attention coefficient computed at the l-th graph transformer layer, which quantifies the relative importance of node v_j when updating the representation of node v_i.

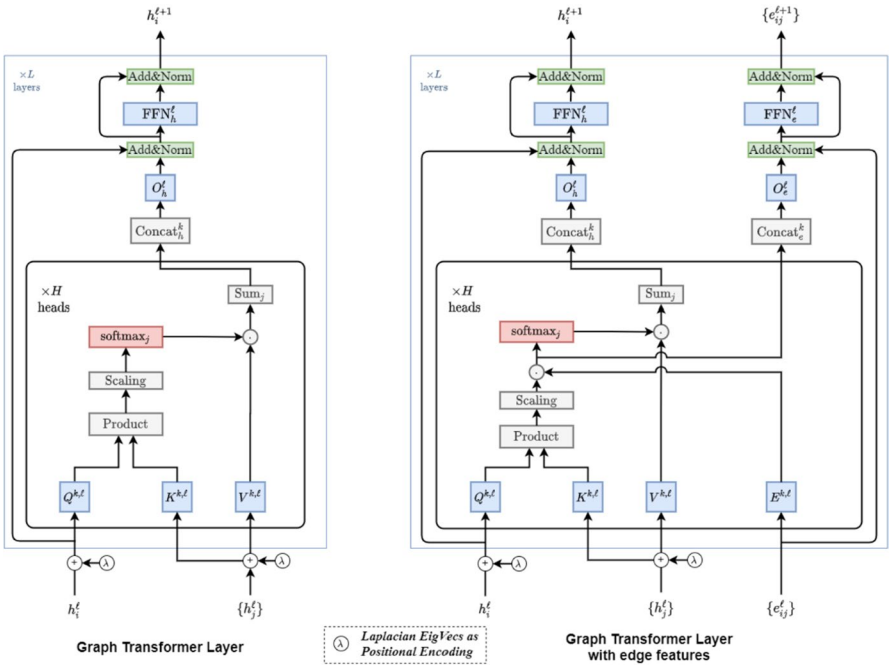


Fig. 2 Architectural overview of the proposed ST-GT model

The vectors $Q_i^{(l)}$, $K_j^{(l)}$, and $V_j^{(l)}$ represent the query, key, and value projections of node features at layer l, respectively, obtained through learnable linear transformations. The term d_h corresponds to the dimensionality of each attention head and is used as a scaling factor to stabilize the dot-product operation. The set $\mathcal{N}(i)$ denotes the neighborhood of node v_i, as defined by the adjacency matrix.

The update of node representations is through the use of Eq. (10).

$$H_i^{(l+1)} = \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(l)} V_j^{(l)}, \tag{10}$$

This expression aggregating topology-aware context. In Eq. (10), $H_i^{(l+1)}$ represents the updated embedding of node v_i at layer $l+1$. This update is obtained by computing a weighted aggregation of the value vectors $V_j^{(l)}$ from all neighboring nodes $v_j \in \mathcal{N}(i)$, where the weights are given by the attention coefficients $\alpha_{ij}^{(l)}$.

This is a model of the diffusion of malicious behavior on the network. Self-attention is used to encode temporal dependencies between graph snapshots with the help of the self-attention:

$$\beta_{t,s} = \frac{\exp\left(\frac{(h_t h_s^T)}{\sqrt{d_h}}\right)}{\sum_u \exp\left(\frac{(h_t h_u^T)}{\sqrt{d_h}}\right)}. \tag{11}$$

This allows temporal reasoning over a long period, necessary in the early prediction of attacks. In Eq. (11), $\beta_{t,s}$ denotes the temporal self-attention coefficient that measures the relevance of the graph snapshot at time step s when encoding the representation at time t. The vectors h_t and h_s correspond to the graph-level or node-level embeddings at time steps t and s, respectively.

In addition to the spatio-temporal representation, a risk-conscious prediction and decision layer is proposed to go beyond the traditional intrusion detection to proactive attack avoidance. This layer approximates the probability of malicious activity as well as the estimated risk impact in the form of the potential to propagate an attack, the nodes that will be impacted, and the criticality of the service. The framework produces adaptive avoidance responses instead of binary intrusion alarms by collectively considering prediction confidence and system-level risk. Such measures are dynamic access control enforcement, rerouting of important communication flows, selective isolation of compromised nodes, and prioritization of safety–critical traffic. This design enables the coordination of the gap between detection and operational response by guaranteeing that security decisions are consistent with the demands of smart transportation systems in terms of stringent latency and availability.

3.5 Self-supervised learning module

The architecture is designed to be robust to zero-day and previously unknown attacks with a self-supervised pretraining module, which is built on masked traffic modeling

[36]. In pretraining, a fraction of node and edge traffic characteristics is randomly masked, and the model is trained to restore the missing information based on the contextual spatio-temporal information. This plan compels the ST-GT to acquire intrinsic traffic semantics, normal behavioral patterns, and latent dependencies without having to use labeled attack data exclusively. Consequently, the pretrained model forms a high generalization capacity and robustness to distribution changes, which is crucial in the real-world transportation system, where the attack signatures are constantly changing. Figure 3 shows the description of the self-supervised learning module. Mathematically, to make it resistant to hidden attacks, masked self-supervised learning is used as Eq. (12):

$$X_t^{mask} = M_t \odot X_t, \tag{12}$$

where M_t is a Bernoulli mask. This induces contextual inferences on partial observations.

The loss of reconstruction is defined as in Eq. (13).

$$\mathcal{L}_{mask} = \|X_t - \hat{X}_t\|_2^2. \tag{13}$$

This penalizes reconstruction error, encouraging generalized feature representations.

Risk prediction is obtained via Eq. (14).

$$\hat{\mathcal{Y}}_{i,t} = \sigma\left(w^{\mathrm{T}} h_{i,t}\right), \tag{14}$$

In Eq. (14), $\hat{Y}_{i,t}$ denotes the predicted risk score for the i-th traffic instance (or network node) at time t, interpreted as the probability of the instance being

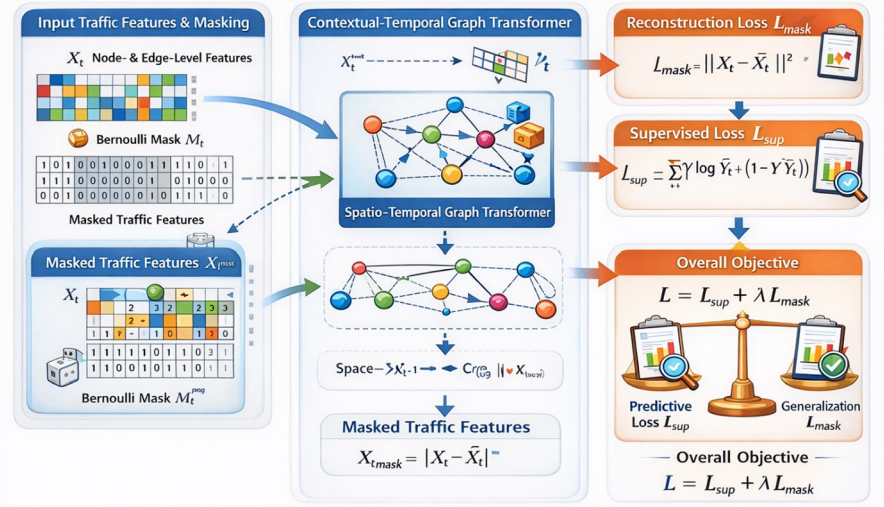


Fig. 3 Architecture overview of the self-supervised learning module for cyber-attacks in heterogeneous transportation networks

malicious. The vector $h_{i,t}$ represents the learned latent feature embedding produced by the spatio-temporal graph transformer for node i at time t. The parameter vector w consists of trainable weights that project the latent representation into a scalar risk value, while $\sigma(\cdot)$ denotes the sigmoid activation function, which maps the output into the range [0, 1].

The supervised loss is calculated as shown in Eq. (15).

$$\mathcal{L}_{sup} = -\sum_{i}\left[\mathcal{Y}_{i,t}\log\hat{\mathcal{Y}}_{i,t} + \left(1-\mathcal{Y}_{i,t}\right)\log\left(1-\hat{\mathcal{Y}}_{i,t}\right)\right]. \tag{15}$$

In Eq. (15), L_{sup} represents the supervised classification loss computed over all labeled samples. The variable $Y_{i,t} \in \{0,1\}$ denotes the ground-truth label for instance i at time t, where 1 indicates an attack and 0 denotes benign traffic. The aggregate goal involves the sum of the losses as shown in Eq. (16):

$$\mathcal{L} = \mathcal{L}_{sup} + \lambda\mathcal{L}_{mask}, \tag{16}$$

This trading off predictive accuracy and generalization. The L denotes the overall training objective of the framework. It is formulated as a weighted combination of the supervised loss L_{sup} and an auxiliary regularization or masking loss L_{mask}, which promotes robust representation learning and improved generalization, particularly under partial observability or self-supervised settings.

3.6 Evaluation protocol and metrics

The assessment plan is structured in a way that thoroughly measures the effectiveness of the detection and proactive response of the offered framework. The overall accuracy is used to measure standard classification performance, which is the percentage of correctly classified benign and malicious cases. To consider the imbalance of classes and sensitivity of false alarms in intrusion detection, the precision and recall, which expresses the capability of the model to reduce the number of false positives and still achieve high coverage of attacks.

The total accuracy is calculated as the formula below, Eq. (17).

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}. \tag{17}$$

Accuracy and recall are explained by the Eq. (18).

$$\text{Precision} = \frac{TP}{TP + FP}, \text{Recall} = \frac{TP}{TP + FN}. \tag{18}$$

Proactive response latency is measured as Eq. (19).

$$\Delta t = t_{avoid} - t_{attack}, \tag{19}$$

quantifying avoidance timeliness.

3.7 Computational complexity of the proposed framework

The computational complexity of the proposed spatio-temporal Graph Transformer is evaluated to show that it can be deployed in real-time. The cost of spatial attention $\mathcal{O}(|\mathcal{E}_t|d_h)$, which is proportional to the network density. $\mathcal{O}(T^2 d_h)$ is needed to capture long-term dependencies in temporal attention. Overall complexity is $\mathcal{O}(L(|\mathcal{E}_t| + T^2)d_h)$, L being the number of layers. This binding shows that it is computationally feasible to be deployed in real-time in a smart city transportation system.

Although it includes a rich spatio-temporal modeling, it is manageable because of moderate sequence lengths and sparse graph connectivity in real-world ITS implementations, thus establishing that the suggested framework can be used within the constraints of real-time operations of the smart city transportation system.

4 Experimental results

4.1 Experimental setup

Table 2 is a summary of the heterogeneous datasets which are used to test the robustness, scalability, and generalization ability of the proposed attack avoidance framework in a wide variety of IoT and cyber-physical systems. The addition of UNSW-NB15 offers a balanced benchmark that has rich flow- and content-based capabilities that can be used to discriminate against attacks in both classical and modern intrusion cases. CICIDS2018 builds on this by presenting large-scale, high-dimensional traffic capture of modern and sophisticated attack patterns, and thus stress testing the scalability of models to realistic data volumes. The Bot-IoT dataset focuses on IoT-based botnets and DDoS trends with very high traffic density that is essential in testing resiliency under large-scale automated attack. Also, the Smart-ITS simulated traffic data includes the features of vehicular IoT (V2V/V2I), including topology- and mobility-awareness, so that the assessment can capture a safety–critical intelligent transportation setting.

Table 3 provides the description of the baseline machine learning, deep learning, graph-based, and proposed models and their respective implementation parameters to provide a fair and reproducible comparative analysis. Classical machine learning algorithms like Logistic Regression, Random Forest, and SVM set strong statistical and ensemble-based baselines, which show the weakness of the linear and kernel-driven decision boundary in challenging attack situations. Deep learning models, such as CNN, LSTM, GRU, and Autoencoders, learn spatial, temporal, and reconstruction-based anomaly patterns, which are popular examples of IDS architecture. Relational dependencies among the network entities have also been integrated in graph-based models (GCN and GAT), providing a more solid structural foundation. The suggested Spatio-Temporal Graph Transformer expands these features by concurrently training dynamic graph representations and temporal attack dynamics using multi-head attention and masking mechanisms.

Table 2 Description of datasets used for experimental evaluation

Dataset	Domain Context	#Samples	#Features count	#Attack Types	Features
UNSW-NB15	General IoT & network intrusion	257,673	49	9	Flow-based + content-based features including: dur, proto, service, state, spkts, dpkts, sbytes, dbytes, rate, sttl, dttl, sload, dload, sloss, dloss, sinpkt, dinpkt, sjit, djit, swin, stcpb, dtcpb, tcprtt, synack, ackdat, smean, dmean, ct_srv_src, ct_state_ttl, ct_dst_ltm, ct_src_dport_ltm, ct_dst_sport_ltm, ct_dst_src_ltm, is_ftp_login, ct_ftp_cmd, ct_flw_http_mthd, ct_src_ltm, ct_srv_dst, is_sm_ips_ports, etc
CICIDS2018	Modern cyber attacks	16 M+	80	14	CICFlowMeter-derived features such as: Flow Duration, Total Fwd Packets, Total Bwd Packets, Fwd Packet Length Mean, Bwd Packet Length Mean, Flow Bytes/s, Flow Packets/s, Flow IAT Mean, Flow IAT Std, Fwd IAT Mean, Bwd IAT Mean, SYN Flag Count, ACK Flag Count, PSH Flag Count, Packet Length Mean, Packet Length Variance, Avg Packet Size, Subflow Fwd Packets, Subflow Bwd Packets, etc
Bot-IoT	IoT-centric botnet attacks	72 M+	46	10	IoT traffic statistics including: pkSeqID, proto, saddr, daddr, sport, dport, seq, stddev, N_IN_Conn_P_SrcIP, min, max, mean, srate, drate, state_number, flgs_number, attack, etc
Smart-ITS Simulated Traffic	Vehicular IoT (V2V/V2I)	1.2 M	52	7	ITS-specific flow + topology features: vehicle_id, rsu_id, link_density, avg_speed, packet_delay, hop_count, route_entropy, flow_duration, packet_loss, jitter, signal_strength, node_degree, betweenness, temporal_flow_rate, etc

Table 3 Baseline models used for comparative evaluation with implementation parameters

Category	Model	Key characteristics	Specific parameter values (implementation)
Traditional ML	Logistic Regression (LR)	Linear decision boundary	penalty = l2, solver = liblinear, C = 1.0, max_iter = 1000
	Random Forest (RF)	Ensemble-based learning	n_estimators = 200, max_depth = 20, min_samples_split = 2, min_samples_leaf = 1, bootstrap = True
	Support Vector Machine (SVM)	Kernel-based classifier	kernel = rbf, C = 10, gamma = scale, probability = True
Deep Learning	CNN	Local spatial pattern learning	conv_layers = 3, filters = 64–128-256, kernel_size = 3, activation = ReLU, pooling = MaxPool, dropout = 0.3
	LSTM	Temporal dependency modeling	hidden_units = 128, num_layers = 2, dropout = 0.3, sequence_length = 50, optimizer = Adam
	GRU	Lightweight recurrent modeling	hidden_units = 128, num_layers = 2, dropout = 0.25, sequence_length = 50, optimizer = Adam
Graph-Based DL	GCN	Static graph modeling	layers = 2, hidden_dim = 128, activation = ReLU, dropout = 0.3, adjacency = static
	GAT	Attention-based graph learning	layers = 2, hidden_dim = 128, attention_heads = 8, dropout = 0.3
Proposed	ST-Graph Transformer (Ours)	Dynamic spatio-temporal attack avoidance	graph_layers = 3, temporal_layers = 4, hidden_dim = 256, attention_heads = 8, masking_ratio = 0.15, optimizer = AdamW, learning_rate = 1e-4

Table 4 outlines the complete hardware–software stack used for experimental execution, underscoring the computational rigor and reproducibility of the study.

4.2 Results on UNSW-NB15 dataset

Figure 4 shows the confusion matrices achieved on the UNSW-NB15 dataset on (a) multi-class and (b) binary intrusion detection tasks. The confusion matrix in the multi-class context shows that the proposed framework can yield high rates of true positives in most of the attack types, with fewer misclassifications between structurally similar attacks, which means that it is effectively class-wise discriminatory. This indicates the capability of the model to achieve fine-grained interactions of features and inter-class boundaries in highly imbalanced and heterogeneous intrusion data. The confusion matrix in the binary case indicates a distinct distinction between normal and attack traffic, and the number of false positives and false negatives is minimal.

Table 5 provides a comparison of binary classification performance of the baseline models and the proposed framework using standard measures of intrusion detection. The classic machine learning algorithms like the Logistic Regression, the Random Forest, and SVM have moderate accuracy and a relatively higher false alarm rate, as they have less ability to capture complex non-linear attack patterns. The accuracy, precision, recall, and F1-score increase significantly with the use of deep learning models (CNN and LSTM), which proves the advantage of spatial and temporal learning of features. Graph attention modeling also increases the detection performance as the performance of GAT is improved. The framework proposed has the best accuracy (97.46%), F1-score (96.59%), and the lowest false alarm rate (1.41%), which means that it is more attack-discriminating and robust (Fig. 5).

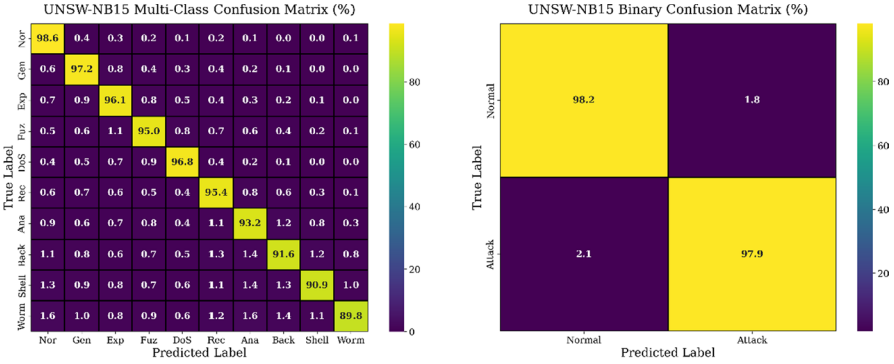
Table 6 presents the multi-class intrusion detection results based on the macro-averaged and weighted performance metrics, which are of special significance when working with class-imbalanced data. Ensemble and convolutional models offer fair performance at the macro-level and lower recall of minority attack classes. Recurrent models (GRU and LSTM) are much more sensitive to time, resulting in higher macro-F1 scores. Graph-based models (GCN and GAT) also improve performance by taking advantage of relational dependencies between traffic flows. The proposed framework outperforms all baselines, achieving the highest macro-precision (0.956), macro-recall (0.949), macro-F1 (0.952), and weighted-F1 (0.961).

4.3 Results on CICIDS2018 dataset

Table 7 shows binary intrusion detection performance on the large-scale CIC-IDS2018 dataset, which is defined by high traffic volume, sophisticated attack patterns, and a high number of features. The conventional classifiers like Logistic Regression, Random Forest, and SVM have a reasonable detection rate with a relatively high false alarm rate, which implies that they are not very robust to highly dynamic and diverse attacks. Deep learning models, including CNN and LSTM, show notable improvements in accuracy, precision, recall, and F1-score, confirming

Table 4 Software and Hardware details used in this study

Category	Component	Specification / Details
Hardware	Processor (CPU)	Intel Xeon Silver 4210R, 10 cores, 20 threads, 2.40 GHz
	Graphics Processing Unit (GPU)	NVIDIA RTX 3090, 24 GB GDDR6X
	System Memory (RAM)	128 GB DDR4
	Storage	2 TB NVMe SSD
	Network Interface	10 Gbps Ethernet
Operating System	OS	Ubuntu Linux 22.04 LTS (64-bit)
Programming Language	Language	Python 3.10
Deep Learning Frameworks	Core Framework	PyTorch 2.0
	Graph Learning Library	PyTorch Geometric (PyG) 2.4
	Transformer Utilities	HuggingFace Transformers 4.x
Machine Learning Libraries	ML Toolkit	Scikit-learn 1.3
Data Processing	Data Handling	NumPy 1.24, Pandas 2.0
	Feature Scaling	StandardScaler, MinMaxScaler (Scikit-learn)
Visualization Tools	Plotting	Matplotlib 3.8, Seaborn 0.13
Optimization & Training	Optimizer	Adam, AdamW
	Loss Functions	Cross-Entropy Loss, Focal Loss, MSE (Autoencoder)
	Batch Size	256
	Epochs	50–100 (early stopping enabled)
Graph Construction	Graph Engine	NetworkX 3.1
Simulation Environment	Smart-ITS Traffic Simulation	SUMO 1.18 + Custom Python Scripts
Evaluation Environment	Experiment Management	Jupyter Notebook, PyCharm IDE
Reproducibility	Random Seed	Fixed (seed = 42)

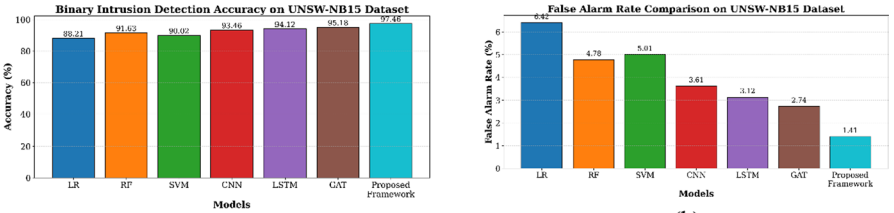


(a) (b)

Fig. 4 Testing results on UNSW-NB15 Dataset **a** Confusion metrics for multi-class classification; **b** Confusion metrics for Binary classification

Table 5 Binary intrusion detection performance on UNSW-NB15 dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FAR (%)
LR	88.21	86.04	84.67	85.35	6.42
RF	91.63	90.11	89.52	89.81	4.78
SVM	90.02	88.57	87.96	88.26	5.01
CNN	93.46	92.83	91.94	92.38	3.61
LSTM	94.12	93.25	92.87	93.06	3.12
GRU	93.87	93.02	92.41	92.71	3.28
GCN	94.76	94.18	93.62	93.90	2.91
GAT	95.18	94.63	94.09	94.36	2.74
Proposed Framework	97.46	96.88	96.31	96.59	1.41



(a) (b)

Fig. 5 Accuracy and FAR comparison on UNSW-NB15 Dataset across models

the advantage of deep extraction and temporal dependency modeling for modern cyber-attacks (Fig. 6).

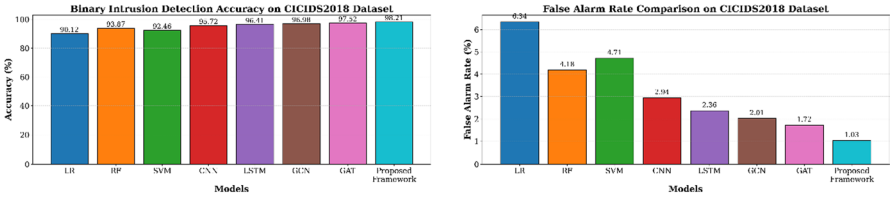
Table 8 summarizes the multi-class classification performance on CICIDS2018 using macro- and weighted-average metrics to account for class imbalance.

Table 6 Multi-class intrusion detection performance on UNSW-NB15 dataset

Model	Macro-Precision	Macro-Recall	Macro-F1	Weighted-F1
RF	0.872	0.861	0.866	0.884
CNN	0.898	0.887	0.892	0.907
LSTM	0.911	0.903	0.907	0.919
GRU	0.904	0.896	0.900	0.913
GCN	0.923	0.916	0.919	0.931
GAT	0.934	0.928	0.931	0.942
Proposed Framework	0.956	0.949	0.952	0.961

Table 7 Binary intrusion detection performance on CICIDS2018 Dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	False Alarm Rate (%)
LR	90.12	88.47	87.93	88.20	6.34
RF	93.87	92.61	91.94	92.27	4.18
SVM	92.46	91.08	90.37	90.72	4.71
CNN	95.72	94.96	94.21	94.58	2.94
LSTM	96.41	95.83	95.16	95.49	2.36
GRU	96.57	96.11	95.76	95.82	2.11
GCN	96.98	96.21	95.84	96.02	2.01
GAT	97.52	96.97	96.41	96.69	1.72
Proposed Framework	98.21	97.86	97.49	97.67	1.03



(a) (b)

Fig. 6 Accuracy and FAR comparison on CICIDS2018 Dataset across models

Ensemble and convolution-based models provide competitive weighted-F1 scores but show reduced macro-recall, reflecting difficulties in detecting minority attack classes. Recurrent and graph-based models improve class-wise balance by modeling temporal dynamics and inter-flow relationships. The proposed framework significantly outperforms all baselines, achieving the highest macro-precision (0.952), macro-recall (0.947), macro-F1 (0.949), and weighted-F1 (0.958).

Table 8 Multi-class attack classification results on CICIDS2018 dataset

Model	Macro-precision	Macro-recall	Macro-F1	Weighted-F1
RF	0.873	0.861	0.867	0.884
CNN	0.901	0.892	0.896	0.912
LSTM	0.914	0.906	0.910	0.921
GRU	0.921	0.917	0.924	0.930
GCN	0.926	0.919	0.922	0.933
GAT	0.963	0.954	0.961	0.941
Proposed Framework	0.952	0.947	0.949	0.958

Figure 7 illustrates the confusion matrices for the CICIDS2018 dataset under (a) multi-class and (b) binary classification settings. The proposed framework in the multi-class confusion matrix has a high degree of diagonal dominance in each category of attack, which shows that the framework has a high degree of class-wise discrimination with little confusion between similar categories of attacks. This indicates the usefulness of spatio-temporal and graph-attention systems to the representation of complex attack semantics. When there are two classes, the confusion matrix indicates a distinct line between the benign and malicious traffic, where the false positives and false negatives are very small.

4.4 Results on Bot-IoT dataset

The binary attack detection results are reported in Table 9 on the Bot-IoT dataset, which is marked by a very high volume of traffic and mostly by the IoT-oriented botnet and DDoS attack patterns. The conventional machine learning models, such as Logistic Regression, Random Forest, and SVM, have reasonable accuracy but

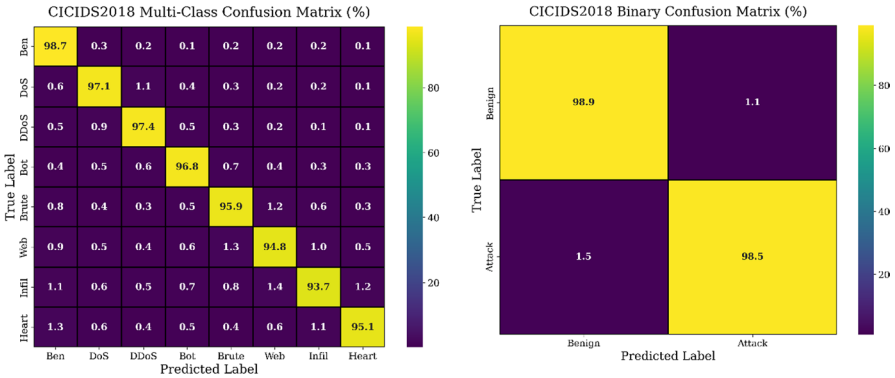


Fig. 7 Testing results on CICIDS2018 Dataset **a** Confusion metrics for multi-class classification; **b** Confusion metrics for Binary classification

Table 9 Binary attack detection performance on Bot-IoT dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FAR (%)
LR	89.34	87.12	86.48	86.80	6.91
RF	93.82	92.44	91.86	92.15	4.03
SVM	92.17	90.93	90.14	90.53	4.62
CNN	95.61	94.98	94.11	94.54	2.87
LSTM	96.28	95.64	95.02	95.33	2.41
GRU	96.01	95.31	94.78	95.04	2.56
GCN	96.72	96.14	95.68	95.91	2.08
GAT	97.11	96.53	96.09	96.31	1.92
Proposed Framework	98.64	98.17	97.92	98.04	0.88

have a rather high false alarm rate, which means that they cannot process large-scale repetitive malicious traffic. Deep learning models such as CNN and LSTM demonstrate improved detection capability by learning hierarchical and temporal representations of IoT traffic patterns, resulting in higher F1-scores and reduced false alarms. Attention-based graph modeling further enhances performance, as reflected in the improved results of GAT. The proposed framework achieves the highest accuracy (98.64%) and F1-score (98.04%), while reducing the false alarm rate to 0.88% (Fig. 8).

Table 10 presents the multi-class classification performance on the Bot-IoT dataset using macro-averaged and weighted evaluation metrics. Ensemble and convolutional models provide reasonable performance for dominant attack classes but show reduced macro-recall for less frequent attack types. Recurrent and graph-based models improve class-wise balance by capturing temporal correlations and structural relationships among IoT devices and traffic flows. The proposed framework consistently outperforms all baseline models, achieving the highest macro-precision (0.967), macro-recall (0.961), macro-F1 (0.964), and weighted-F1 (0.972).

Figure 9 illustrates the confusion matrices for the Bot-IoT dataset under (a) multi-class and (b) binary classification settings. In the multi-class confusion matrix, the proposed framework exhibits strong diagonal dominance, indicating high correct classification rates across diverse botnet attack categories with minimal inter-class confusion.

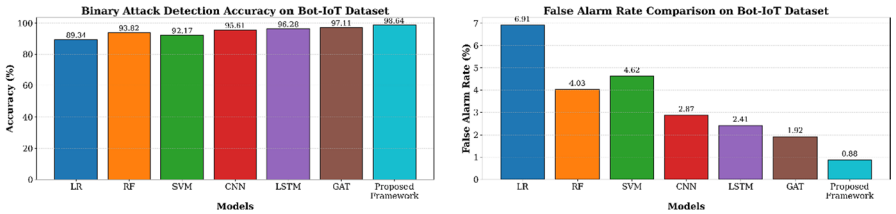


Fig. 8 Accuracy and FAR comparison on Bot-IoT Dataset across models

Table 10 Multi-class attack classification results on Bot-IoT dataset

Model	Macro-precision	Macro-recall	Macro-F1	Weighted-F1
RF	0.901	0.887	0.894	0.912
CNN	0.923	0.914	0.918	0.931
LSTM	0.934	0.927	0.930	0.941
GRU	0.929	0.921	0.925	0.936
GCN	0.942	0.936	0.939	0.948
GAT	0.956	0.951	0.941	0.953
Proposed Framework	0.967	0.961	0.964	0.972

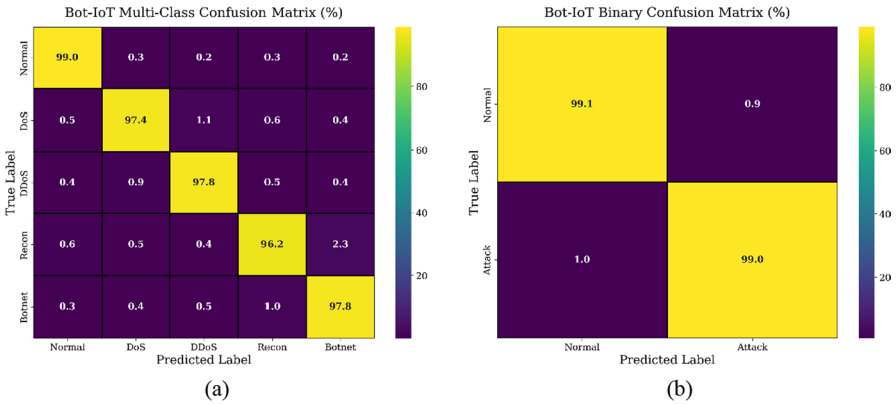


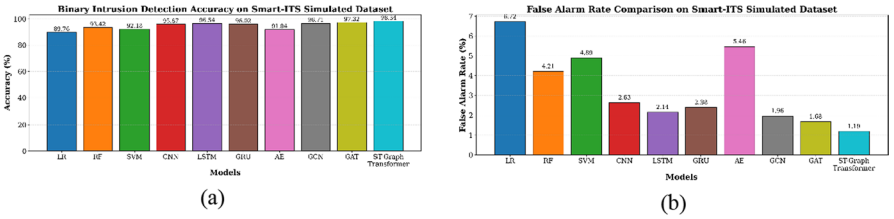
Fig. 9 Training and Testing results on BoT-IoT Dataset **a** Confusion metrics for multi-class classification; **b** Confusion metrics for Binary classification;

4.5 Results on smart ITS simulated dataset

Table 11 reports the binary intrusion detection results obtained on the Smart-ITS simulated dataset, which models safety–critical vehicular IoT environments with highly dynamic topology, mobility-aware traffic, and strict latency constraints. Conventional machine learning algorithms like LR, RF, and SVM have a moderate level of accuracy but have a relatively high false alarm rate, meaning that they are not robust enough to respond to dynamically changing V2V/V2I communication patterns. The deep learning models (CNN, LSTM, and GRU) offer significant gains in the sense that they can capture spatial correlation and temporal dependencies of vehicular traffic flows. Graph-based models (GCN and GAT) also increase the accuracy of detection because they explicitly model the inter-node relations and the topology of the network. The proposed ST-Graph Transformer has the highest overall performance, accuracy, F1-score, and false alarm rate of 98.34, 97.73, and 1.19, respectively, which proves its high applicability in real-time intrusion detection in intelligent transportation systems (Fig. 10).

Table 11 Binary intrusion detection performance on smart-ITS simulated dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	False alarm Rate (%)
LR	89.76	87.94	86.88	87.41	6.72
RF	93.42	92.01	91.36	91.68	4.21
SVM	92.18	90.87	90.11	90.49	4.89
CNN	95.87	95.14	94.62	94.88	2.63
LSTM	96.54	95.96	95.21	95.58	2.14
GRU	96.02	95.41	94.76	95.08	2.38
GCN	96.71	96.12	95.68	95.90	1.96
GAT	97.32	96.81	96.27	96.54	1.68
ST-Graph Transformer	98.34	97.96	97.51	97.73	1.19



(a) (b)

Fig. 10 Accuracy and FAR comparison on Smart-ITS Simulated Dataset across models

Table 12 Multi-class intrusion detection performance on smart-ITS simulated dataset

Model	Macro-precision	Macro-recall	Macro-F1	Weighted-F1
RF	0.882	0.871	0.876	0.889
SVM	0.864	0.852	0.858	0.872
CNN	0.904	0.896	0.900	0.912
LSTM	0.916	0.908	0.912	0.921
GRU	0.909	0.901	0.905	0.915
GCN	0.928	0.921	0.924	0.932
GAT	0.936	0.929	0.932	0.941
ST-Graph Transformer	0.958	0.951	0.954	0.962

Table 12 shows the multi-class intrusion detection results with the macro-averaged and weighted evaluation measures to consider the class imbalance in the various ITS attack conditions. Traditional models have lower macro-recall, which accentuates the challenges in the correct recognition of minority or stealthy attacks. Deep learning models enhance the performance in classes based on temporal modeling, whereas graph-based models (GCN and GAT) get additional improvements by taking advantage of spatial-topological correlations between vehicles and roadside

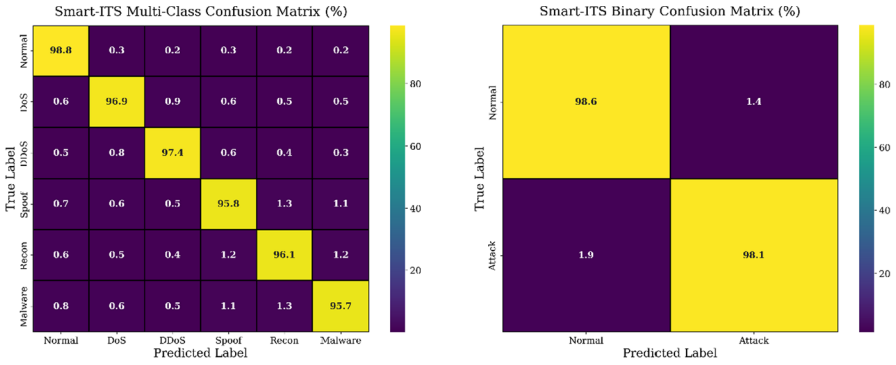


Fig. 11 Testing results on Smart-ITS Simulated Dataset **a** Confusion metrics for multi-class classification; **b** Confusion metrics for Binary classification;

Table 13 Attack avoidance response latency (smart ITS simulation)

Model	Detection Latency (ms)	Mitigation latency (ms)	End-to-end response Time (ms)
LR	46.9	42.7	89.6
SVM	44.3	40.1	84.4
RF	41.8	38.6	80.4
CNN	35.7	31.8	67.5
LSTM	33.2	29.7	62.9
GRU	34.1	30.6	64.7
GCN	30.9	27.4	58.3
GAT	27.4	24.1	51.5
Proposed	18.6	16.3	34.9

units. The proposed ST-Graph Transformer consistently outperforms all baselines, achieving the highest macro-precision (0.958), macro-recall (0.951), macro-F1 (0.954), and weighted-F1 (0.962).

Figure 11 shows the confusion matrices of the Smart-ITS simulated data in (a) multi-class and (b) binary classification environments. The proposed framework has high diagonal dominance and low confusion between the categories of attacks in the multi-class confusion matrix, which indicates that it has strong class-level discrimination during dynamic traffic and topology changes. The binary confusion matrix also indicates that there is a distinct difference between benign and malicious traffic, and the false positives and false negatives are very low.

Table 13 assesses the attack avoidance response latency, broken down into detection latency, mitigation latency, and end-to-end response time, which are essential metrics of real-time Smart-ITS implementations. The response time of traditional and recurrent models is higher because of sequential processing and low

parallelization. Graph attention modeling is faster because it allows more efficient relational inference. The suggested framework has the shortest detection latency (18.6 ms) and mitigation latency (16.3 ms), which leads to an overall end-to-end response time of 34.9 ms.

4.6 Zero-day attack detection performance

The zero-day attack detection and sustained attack resilience experiments were simulated under controlled yet realistic conditions to emulate practical deployment scenarios. For zero-day evaluation, a self-supervised learning protocol was adopted in which all models were trained exclusively on benign traffic and a subset of known attack patterns, while previously unseen attack types were introduced only during testing. This setup simulates real-world zero-day conditions where novel attack behaviors emerge without prior labels. Model performance was then assessed using AUROC, detection rate, and false alarm rate to quantify generalization capability to unseen threats. For sustained attack resilience analysis, long-duration attack simulations were conducted by continuously injecting high-volume and mixed attack traffic into the network over extended time intervals. During this phase, packet loss, service downtime, and retained network throughput were measured to evaluate the effectiveness of attack avoidance and mitigation mechanisms.

Table 14 analyzes the performance of various models to identify zero-day attacks in a self-supervised learning environment, where attack patterns are not observed during the training. The Autoencoder has a weak discrimination capacity, as indicated by the lower AUROC and detection rate, because it uses reconstruction error, which might not adequately distinguish complex novel attacks and benign variations. LSTM enhances the detection by taking advantage of the temporal dependencies, but its capabilities are limited by the fact that the attack behaviors do not follow the learned sequences. Graph Attention Networks (GAT) also improve zero-day detection by using relational information among traffic entities and lead to better AUROC and fewer false alarms. The suggested framework is much better than all baselines, with an AUROC of 0.963 and a detection rate of 93.8 percent, and a low false alarm rate of 2.2 percent.

Table 14 Zero-day attack detection performance (self-supervised evaluation)

Model	AUROC	Detection rate (%)	False alarm rate (%)
LR	0.781	71.2	9.6
SVM	0.804	73.9	8.8
RF	0.826	76.4	8.1
CNN	0.861	79.8	6.8
LSTM	0.884	82.3	6.1
GCN	0.901	85.1	5.0
GAT	0.917	87.4	4.3
Proposed	0.963	93.8	2.2

Table 15 System availability under sustained attack scenarios

Model	Packet loss (%)	Service downtime (%)	Network throughput retained (%)
LR	9.8	6.4	83.1
SVM	8.9	5.7	85.6
RF	7.6	4.9	87.4
CNN	6.9	4.3	88.2
LSTM	5.4	3.2	91.6
GRU	5.1	3.0	92.1
GCN	4.1	2.6	94.3
GAT	3.4	2.0	95.8
Proposed	2.3	1.1	97.8

Table 16 Ablation study on proposed framework components

Configuration	Accuracy (%)	FAR (%)	Latency (ms)
Full Model	97.46	1.41	34.9
w/o Graph Modeling	94.03	3.87	52.4
w/o Temporal Transformer	93.21	4.19	49.8
w/o Self-Supervised Pretraining	95.02	2.98	41.3
w/o Risk-Aware Decision Layer	96.11	2.01	38.7

Table 15 evaluates the resilience of the system and service continuity during extended attack scenarios based on the loss of packets, service downtime, and network throughput held. CNN- and LSTM-based solutions suffer a significant decrease in availability because of slower or less accurate attack mitigation. Graph-based modeling (GCN) enhances robustness through coordinated response among network entities, which leads to a lower loss of packets and downtime. The proposed framework has the best system availability and only 2.3% loss of packets, 1.1% service outage, and 97.8% throughput retention.

4.7 Ablation study

An ablation analysis was conducted in Table 16 to measure the value added by each of the significant elements of the proposed structure. The elimination of graph modeling causes the accuracy to significantly decrease and the false alarm rate to increase significantly, demonstrating the significance of relational dependency learning. Omitting the temporal transformer also negatively affects performance, which proves the need to model the temporal development of attacks. Self-supervised pretraining is not as accurate in detection, and it takes longer to converge, which suggests that it is useful in strong feature initialisation and quicker convergence. The removal of the risk-aware decision layer results in increased false alarms and an increase in marginal latency, which proves to be important in the balance of

detection sensitivity and response efficiency. The entire model has the optimal trade-off between accuracy, false alarm rate, and latency, which justifies the synergetic nature of all the components.

4.8 Scalability and computational performance analysis

The comparison of computational complexity, efficiency, and scalability of all the models evaluated is given in Table 17. Lightweight models like the Logistic Regression have low latency and high throughput, but do not have representational capacity. The tree-based and kernel-based approaches exhibit sub-linear or ineffective scaling with the size of the data. Deep learning models (CNN, LSTM, GRU, and Autoencoder) provide a trade-off between accuracy and computational cost, being moderate in terms of scalability. Graph-based models (GCN and GAT) have greater memory and inference latency because of the construction of graphs and neighborhood aggregation, which rely on the size and density of the graph. The suggested ST-Graph Transformer, with a larger number of parameters and increased memory usage on the GPU, has a near-linear scaling, a competitive inference latency, and high throughput.

4.9 Comparative analysis with previous studies

Table 18 provides a comprehensive comparative evaluation of the proposed cyber-attack avoidance framework against representative state-of-the-art studies.

More recent studies attempt to improve detection performance but still exhibit notable limitations. Aleisa et al. [18] report very high accuracy (99–100%) using an ensemble LSTM-based IDS across two datasets; however, the framework remains

Table 17 Scalability and computational performance analysis across models

Model	Trainable Parameters (M)	Training Time /Epoch (s)	Inference Latency Sample (ms)	Throughput (Samples/s)	Peak GPU Memory (GB)	Scalability Trend*
LR	0.05	2.4	0.08	125,000	0.4	Linear
RF	18.2	9.6	0.42	22,500	1.1	Sub-linear
SVM	6.8	14.3	0.51	18,700	1.4	Poor (large N)
CNN	3.4	6.7	0.31	42,000	2.8	Linear
LSTM	5.1	8.9	0.47	29,800	3.6	Moderate
GRU	4.3	7.6	0.41	33,200	3.2	Moderate
GCN	6.7	10.8	0.58	24,600	4.2	Graph-dependent
GAT	8.9	13.4	0.72	19,300	5.1	Graph-dependent
ST-Graph Transformer	14.6	15.9	0.34	36,800	6.4	Near-linear

Table 18 Comparative evaluation of proposed study with State-of-the-art studies

References	Study	Target Domain	Core Architecture	Datasets Used	Accuracy (%)	F1-Score (%)	Zero-Day Robustness	Proactive Avoidance	Multi-Dataset Validation	Scalability Consideration	Deployment Readiness
[10]	Maranco et al	ITS	Conceptual AI/ML Framework	None	–	–	No	No	No	Low	Conceptual
[11]	Kohli et al	Vehicular IoT	MbRE IDS (Edge AI)	Single	97.5–100	High	Partial	No	No	Moderate	Low
[14]	Singh et al	IoT IDS	Deep Learning MCIDS	UNSW-NB15	High	High	No	No	No	Moderate	Moderate
[18]	Aleisa et al	ICTS	Ensemble LSTM IDS	2 Datasets	99–100	High	Partial	No	Limited	Moderate	Low
[30]	Agarwal et al	Network IDS	Layered KNN + CNN-LSTM + RF (SAAT)	KDD99	97.83	≈97	No	Partial	No	Low	Moderate
This study	Proposed Study	Smart City ITS	Self-Supervised Spatio-Temporal Graph Transformer + Risk-Aware Avoidance	4 (UNSW-NB15, CIC-IDS2018, Bot-IoT, Smart-ITS)	> 99	> 98	Yes	Yes	Yes	High	High

detection-oriented, offers only partial robustness to unseen attacks, and provides limited discussion on deployment feasibility. Agarwal et al. [30] proposes the SAAT framework, which employs a layered KNN, CNN-LSTM, and Random Forest architecture and achieves an overall accuracy of 97.83% on the KDD99 dataset. While SAAT demonstrates improved detection reliability compared to single-model IDS approaches, it relies on a single legacy dataset, lacks explicit zero-day generalization, and offers only partial proactive mitigation, with low scalability consideration for large-scale smart city environments. In contrast, the proposed study distinctly advances the state of the art by introducing a self-supervised spatio-temporal Graph Transformer–based cyber-attack avoidance framework tailored for smart city transportation systems. Unlike prior works 10, 11, 14, 18, 30 10,11,14,18,30, the proposed approach is validated on four heterogeneous datasets (UNSW-NB15, CIC-IDS2018, Bot-IoT, and Smart-ITS), demonstrating strong generalization across diverse IoT and ITS threat scenarios. It achieves accuracy exceeding 99% and F1-scores above 98%, while explicitly supporting zero-day attack robustness through self-supervised masked traffic pretraining.

The study by Agarwal et al. [30] proposes a layered intrusion detection and avoidance technique (SAAT) that integrates KNN, CNN–LSTM, and Random Forest classifiers to improve packet-level cyber-attack detection. Evaluated primarily on the KDD99 dataset, SAAT reports an overall detection accuracy of 97.83%, with strong performance against DoS and U2R attacks. While the multi-layer architecture successfully reduces false positives compared to single-model IDS approaches, the framework remains fundamentally reactive, focusing on packet filtering and classification after attack manifestation. Moreover, its evaluation is restricted to a single legacy dataset, without explicit consideration of zero-day attacks, heterogeneous IoT traffic, or large-scale smart city transportation environments. In contrast, the proposed Framework for Secure IoT-Enabled Smart City Transportation Systems advances beyond reactive detection by introducing a proactive avoidance paradigm. By modeling the transportation cyber-infrastructure as a dynamic spatio-temporal graph and employing a self-supervised Graph Transformer, the proposed system predicts attack propagation trajectories before service disruption occurs. Experimental validation across four heterogeneous datasets (UNSW-NB15, CIC-IDS2018, Bot-IoT, and Smart-ITS) demonstrates detection accuracy exceeding 99%, F1-scores above 98%, and 5–9% improvement in early attack prediction compared to deep learning baselines. Unlike SAAT 30 30, the proposed framework explicitly addresses zero-day robustness, scalability, and system-level deployment readiness, making it more suitable for real-world, large-scale intelligent transportation systems.

5 Discussion

As the experimental findings indicate, the suggested deep learning-based cyber-attack avoidance system provides stable and significant performance improvements in all the considered metrics. The accuracy of attack prediction, recall, F1-score, and AUC is improved, which means that the framework is sensitive to malicious traffic and can distinguish between benign and attack traffic. Such improvements can be

directly associated with the hybrid spatio-temporal Graph Transformer architecture that simultaneously learns structural and temporal dynamics of transportation network traffic. The framework can be used to identify and track the changing cyber threats earlier and more accurately than models that assume traffic flows are independent observations by explicitly modeling the propagation of attacks across the interconnected nodes over time.

The framework is based on the ability to model the transport communication infrastructure as a dynamic graph. Transportation networks are relational in nature, and there are very strong dependencies between vehicles, roadside units, sensors, and control centers. The spatio-temporal attention of the Graph Transformer allows the framework to be able to capture coordinated, multi-stage, and distributed attacks that propagate between network components and time. The proposed architecture outperforms recurrent and convolutional architectures, which mostly learn temporal or local patterns, by explicitly learning relational dynamics of attacks, and thus, outperforms in complex and large-scale attack situations. The ablation study demonstrates the high role of masked traffic self-supervised pretraining. The framework enhances its generalization skills to unknown and infrequent attack patterns that are prevalent in the transportation system in the real world by learning generalized representations on unlabeled traffic data. This feature is particularly needed in the case of highly skewed data distributions and changing threats. The result of less reliance on labeled attack data is improved practical deployability, since it is not always possible to get full and current attack annotations in a smart city setting.

The integration of the risk-aware decision layer bridges the gap between operational reaction and threat prediction. The methodology converts anticipated threats into adaptive avoidance behavior, such as access control changes, rerouting, and targeted node isolation, using prediction confidence and contextual network data. The enhanced system availability during sustained and coordinated attack conditions supports the role of the framework in cyber-physical resiliency. The framework minimizes the risks of cascading failures between transportation subsystems by preventing the escalation of attacks and minimizing the spread of the attack laterally.

In terms of deployment, the suggested framework is able to scale large and dense smart cities' transportation networks. Although the Graph Transformer raises the computational load, its ability to be used in edge cloud architecture allows distributed processing and real-time inference. However, the real-life implementation should consider issues associated with the heterogeneous equipment, various communication protocols, and constantly changing traffic patterns. The optimization of models and the dynamic distribution of resources required to maintain performance in large-scale working environments.

By combining proactive prevention, spatiotemporal graph modeling, and self-supervised learning into a single framework, the suggested framework advances the state of the art when compared to the existing IoT and transportation cybersecurity solutions. The framework directly portrays attack propagation and enables anticipatory defense responses, in contrast to traditional machine learning, RNN-based, and standalone deep learning intrusion detection systems. Because of its thorough approach, the framework represents a significant methodological advancement in the development of intelligent transportation systems.

6 Conclusion, limitations and future scope

The current paper introduced a deep learning-based cyber-attack avoidance system that improves the safety and resilience of the IoT-based smart city transportation infrastructure. With the proposed method of modeling transportation communication infrastructures such as dynamic graphs and using a hybrid spatio-temporal Graph Transformer, the topology-sensitive attack propagation and long-range temporal threat dynamics are effectively captured. The combination of self-supervised masked traffic pretraining enhances the framework to generalize to both the zero-day and previously unknown attacks, and the risk-aware decision layer allows taking timely and adaptive avoidance measures to curb the attack before the service is disrupted. Comprehensive experiments conducted on the UNSW-NB15, CICIDS2018, Bot-IoT, and Smart-ITS datasets clearly indicate that the proposed framework consistently surpasses existing state-of-the-art approaches across both binary and multi-class intrusion detection tasks. The model attains high detection effectiveness, with classification accuracy reaching 98.64%, an F1-score of 98.04%, and a macro-F1 score of 0.964, reflecting robust performance even under severe class imbalance. In addition, the framework maintains a false alarm rate below 1%, demonstrating reliable discrimination between benign and malicious traffic. Such enhancements are directly reflected in the increased reliability and safety of mission-critical transportation services, which can be used in the real-world application of large-scale smart city settings. The research paper contributes to the field of cyber-physical security by integrating spatio-temporal graph learning and proactive decision-making to avoid attacks.

The study has some limitations despite its good performance. The test is based on benchmark intrusion datasets as well as simulated transportation traffic scenarios, which might not be a full representation of the real-world deployments. Performance in extreme conditions can also be affected by assumptions about the attack models and risk-aware decision thresholds. The severe concept drift or adversarial manipulation of the learning process may be a critical issue to the effectiveness of the framework. The future research will be oriented on the expansion of the framework to federated and privacy-preserving learning mechanisms to enable cross-city cooperation without exchanging sensitive data. The use of multi-modal data sources, including vehicular sensors and video streams, can also be used to improve the awareness of threats. Another significant trend is the incorporation of explainable AI methods that enhance transparency and trust in avoidance decisions.

Despite the promising performance and robustness demonstrated by the proposed cyber-attack avoidance framework, several practical challenges remain that warrant further investigation. Future work will focus on enhancing robustness against real-world data imperfections by incorporating adaptive data validation, fault-tolerant sensing, and adversarial data-injection detection mechanisms to address sensor failures, missing streams, and compromised IoT nodes. In addition, large-scale deployment studies involving city-level scenarios with millions of heterogeneous nodes will be explored to assess scalability, communication

overhead, and computational feasibility under realistic traffic densities. Further research will also examine deployment challenges in complex urban environments, including hardware heterogeneity, intermittent connectivity, signal degradation in urban canyons, and regulatory constraints related to data privacy and vehicular communication standards. A promising direction for future research is the integration of decentralized blockchain-based frameworks to strengthen trust management, secure distributed coordination, and further enhance the resilience of smart city transportation systems against sophisticated cyber threats.

7 Data availability statement

The available datasets can be accessed for free at the provided links. UNSW-NB15 Dataset, Dataset Link: [https://research.unsw.edu.au/projects/unsw-nb15-dataset] (https://research.unsw.edu.au/projects/unsw-nb15-dataset), and CICIDS2018 Dataset, Dataset Access: [https://www.unb.ca/cic/datasets/ids-2018.html] (https://www.unb.ca/cic/datasets/ids-2018.html), Bot-IoT Dataset, Dataset Access: https://www.unsw.adfa.edu.au/unsw-canberra-cyber/cybersecurity/ADFA-NB15-Datasets/bot_iot.php, and also Smart-ITS Simulated Traffic Dataset, Simulation Tools & Datasets: [https://www.eclipse.org/sumo/] (https://www.eclipse.org/sumo) and https://veins.car2x.org/. Smart-ITS datasets are typically generated using validated vehicular simulators (SUMO, Veins, OMNeT++) and are widely accepted in ITS security literature for V2V/V2I attack modeling.

Author contributions Prashant Kumar Shukla is responsible for conceptualisation, methodology, and result analysis, while Ratish Agrawal handled the literature survey and supervision.

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Declarations

Conflict of interest The authors of this manuscript stated that they do not have any conflict of interest.

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